

SAT Math Mastery · Week 4B

Polynomial, Rational & Radical Equations

Time 150 minutes: 110 min solving + 40 min self-check and error log

Topics factor theorem and remainders · polynomial structure and zeros · rational equations · radical equations and extraneous solutions · equivalent expressions and exponents · literal equations · building expressions

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Use the factor theorem and remainder theorem: $(x - r)$ is a factor $\iff p(r) = 0$; remainder on division by $x - r$ is $p(r)$.
- S2.** Read zeros and factors from tables and graphs; match a graph (crossing, bouncing, end behaviour) to an equation.
- S3.** Solve rational equations by clearing denominators and discard solutions that make a denominator 0.
- S4.** Solve radical equations by isolating the radical and squaring; check every candidate in the original equation.
- S5.** Rewrite and combine rational expressions; simplify expressions with rational exponents and radicals.
- S6.** Rearrange literal equations and formulas to isolate a chosen variable.
- S7.** Use structure: substitution $u = x^2$, symmetric expressions, Vieta for cubics.
- S8.** Build one-variable expressions from geometry or word contexts (area, volume, rates) and solve.

2 · REQUIRED FACTS

- Factor theorem: $(x - r)$ is a factor of $p(x) \iff p(r) = 0$. Remainder theorem: $p(x) \div (x - r)$ leaves remainder $p(r)$.
- A table entry $p(r) = 0$ gives the factor $(x - r)$; a zero that *touches* the axis without crossing has even multiplicity.
- A rational expression is undefined where its denominator is 0; these values are excluded from every solution set.
- Squaring both sides of an equation can create *extraneous* solutions. Always substitute back, and remember $\sqrt{} \geq 0$.
- $a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$, $a^{-n} = \frac{1}{a^n}$, $a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$.
- To isolate a variable, undo operations in reverse order; if the target appears twice, gather those terms and factor it out.

3 · REQUIRED FORMULAS

$$p(x) = (x - r)q(x) + p(r), \quad (x - r) \mid p(x) \iff p(r) = 0$$

$$x^3 + bx^2 + cx + d: \quad r_1 + r_2 + r_3 = -b, \quad r_1 r_2 + r_1 r_3 + r_2 r_3 = c, \quad r_1 r_2 r_3 = -d$$

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}, \quad \sqrt{x} = x - k \Rightarrow x = (x - k)^2 \text{ (then check } x - k \geq 0), \quad r^2 + s^2 + t^2 = (r + s + t)^2 - 2(rs + rt + st)$$

4 · MINI THEORY SHEET

Factor and remainder. To test a candidate factor $(x - r)$, evaluate $p(r)$; if it is 0 the factor works. If the remainder is given, set $p(r) = \text{remainder}$ and solve for the constant.

Rational equations. (1) Note the excluded values. (2) Multiply every term by the common denominator. (3) Solve. (4) Reject any solution equal to an excluded value. If every solution is rejected, there is *no solution*.

Radical equations. (1) Isolate one radical. (2) Square both sides (expand $(x - k)^2$ fully). (3) Solve. (4) Check each candidate in the *original* equation: the radical side must be ≥ 0 .

Equivalent expressions. Put fractions over a common denominator; write radicals as fractional exponents, combine, then convert back. Test a choice by plugging in a convenient number.

Literal equations. Treat other letters as numbers. Clear fractions first, collect the target variable on one side, factor, divide. Roots and squares are undone by squaring and square-rooting.

Graph to equation. Zeros give factors. A graph that bounces at a zero has a squared factor. End behaviour fixes the sign of the leading coefficient. A vertical asymptote at $x = c$ means $(x - c)$ is in the denominator; the horizontal asymptote is the ratio of leading coefficients.

Speed shortcuts.

Situation	Shortcut
“is $(x - r)$ a factor?”	compute $p(r)$ only: no division
remainder on division by $(x - r)(x - s)$	remainder is $ax + b$; use $p(r)$ and $p(s)$ as two equations
zeros of a monic cubic, constant d	product of zeros $= -d$; sum $= -b$
$x^4 + bx^2 + c$	substitute $u = x^2$; real x needs $u \geq 0$
$\sqrt{x} + \frac{1}{\sqrt{x}} = s$	square: $x + \frac{1}{x} = s^2 - 2$
radical equation answer choices	plug in the choices; the extraneous one fails
rational equation with a parameter	solve for x , then ask when x hits the excluded value
rewriting with exponents	test with $x = 2$ or $x = 4$ on both sides

5 · GUIDED EXAMPLES

Example 1 — Factor theorem to find a constant

$x + 3$ is a factor of $p(x) = x^3 + kx^2 - 4x + 3$. Find k .

$x + 3 = x - (-3)$, so $p(-3) = 0$: $-27 + 9k + 12 + 3 = 0 \Rightarrow 9k = 12 \Rightarrow k = \frac{4}{3}$.

Example 2 — Rational equation and an excluded value

Solve $\frac{x^2}{x - 4} = \frac{16}{x - 4}$.

Excluded value: $x = 4$. Multiply by $(x - 4)$: $x^2 = 16 \Rightarrow x = \pm 4$. The solution $x = 4$ makes the denominator 0, so reject it. The only solution is $x = -4$.

Example 3 — Radical equation and an extraneous solution

Solve $\sqrt{x + 6} = x$.

Square: $x + 6 = x^2 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x - 3)(x + 2) = 0$. Check $x = 3$: $\sqrt{9} = 3$ ✓. Check $x = -2$: $\sqrt{4} = 2 \neq -2$, extraneous. The only solution is $x = 3$.

6 · PRACTICE SET — 48 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 + x^2 - 10x + 8$ has exactly one of the following as a factor. Which one?

- (A) $x + 2$ (B) $x - 1$
(C) $x + 1$ (D) $x - 4$

2. EASY · NO CALCULATOR NEEDED

When $f(x) = x^3 - 3x + 5$ is divided by $x - 2$, what is the remainder?

3. EASY · NO CALCULATOR NEEDED

Which of the following expressions is equivalent to $\frac{2x^2 - 18}{x - 3}$ for all $x \neq 3$?

- (A) $2x + 3$ (B) $x + 3$
(C) $2x + 6$ (D) $2x - 6$

4. EASY · NO CALCULATOR NEEDED

What is the solution to the equation $\sqrt{3x + 4} = 7$?

5.

EASY · NO CALCULATOR NEEDED

The volume of a cone is $V = \frac{1}{3}\pi r^2 h$. Which of the following gives h in terms of V and r ?

(A) $h = \frac{3V}{\pi r^2}$

(B) $h = \frac{V}{3\pi r^2}$

(C) $h = 3V\pi r^2$

(D) $h = \frac{3V}{\pi r}$

6.

MEDIUM · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 + kx^2 - 5x - 6$, where k is a constant, has $x - 2$ as a factor. What is the value of k ?

7.

MEDIUM · NO CALCULATOR NEEDED

The table gives selected values of a polynomial function p .

x	-3	-1	2	4
$p(x)$	5	0	9	-2

Which of the following must be a factor of $p(x)$?

(A) $x + 3$

(B) $x - 4$

(C) $x - 1$

(D) $x + 1$

8.

MEDIUM · NO CALCULATOR NEEDED

What is the solution set of the equation $\frac{x}{x-4} = \frac{4}{x-4} + 3$?

- (A) $\{-4\}$ (B) $\{2\}$
(C) No solution (D) $\{4\}$

9.

MEDIUM · NO CALCULATOR NEEDED

What is the solution to $\sqrt{x+7} = x-5$?

10.

MEDIUM · NO CALCULATOR NEEDED

Which of the following is equivalent to $\frac{\sqrt[3]{x^5}}{x}$, for $x > 0$?

- (A) $x^{3/2}$ (B) $x^{2/3}$
(C) $x^{5/3}$ (D) $x^{-2/3}$

11.

MEDIUM · NO CALCULATOR NEEDED

The period of a pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$, where L is its length and g is the acceleration due to gravity. Which of the following gives L in terms of T and g ?

(A) $L = \frac{gT^2}{2\pi}$

(C) $L = \frac{4\pi^2}{gT^2}$

(B) $L = \frac{gT}{2\pi}$

(D) $L = \frac{gT^2}{4\pi^2}$

12.

MEDIUM · NO CALCULATOR NEEDED

How many distinct real solutions does the equation $x^4 - 5x^2 - 36 = 0$ have?

(A) 2

(B) 3

(C) 4

(D) 0

13.

MEDIUM · NO CALCULATOR NEEDED

A closed rectangular box has a square base. Its height is 2 centimeters more than the side length of its base, and its volume is 96 cubic centimeters. What is the total surface area, in square centimeters, of the box?

14.

HARD · NO CALCULATOR NEEDED

When the polynomial $p(x)$ is divided by $x - 3$, the remainder is 5. When $p(x)$ is divided by $x + 1$, the remainder is -3 . When $p(x)$ is divided by $(x - 3)(x + 1)$, the remainder is $ax + b$, where a and b are constants. What is the value of $a + b$?

15.

HARD · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 + ax^2 + bx - 12$, where a and b are constants, has zeros at $x = 1$ and $x = -3$. What is the third zero of p ?

(A) -12 (B) -4 (C) 4 (D) 12

16.

HARD · NO CALCULATOR NEEDED

What is the solution to the equation $\frac{x}{x-3} + \frac{2}{x+1} = \frac{12}{x^2 - 2x - 3}$?

17.

HARD · NO CALCULATOR NEEDED

What is the solution to the equation $\sqrt{x+4} - \sqrt{x-1} = 1$?

18.

HARD · NO CALCULATOR NEEDED

Which of the following is equivalent to $\frac{2}{x-1} - \frac{3}{x+2}$?

(A) $\frac{-1}{(x-1)(x+2)}$

(B) $\frac{7-x}{2x+1}$

(C) $\frac{7-x}{(x-1)(x+2)}$

(D) $\frac{1-x}{(x-1)(x+2)}$

19.

HARD · NO CALCULATOR NEEDED

The formula $P = \frac{4Q}{R+Q}$ relates the positive quantities P , Q , and R , where $P < 4$. Which of the following gives Q in terms of P and R ?

(A) $Q = \frac{PR}{4-P}$

(B) $Q = \frac{PR}{P-4}$

(C) $Q = \frac{PR}{P+4}$

(D) $Q = \frac{(4-P)R}{P}$

20.

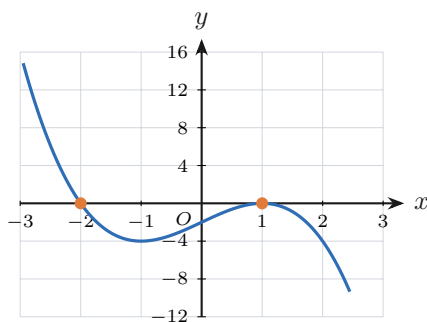
HARD · NO CALCULATOR NEEDED

The polynomial $x^3 - 7x^2 + 14x - 8$ has three real zeros r , s , and t . What is the value of $r^2 + s^2 + t^2$?

21.

HARD · DESMOS

The graph of the polynomial function p is shown in the xy -plane. Which of the following could define $p(x)$?



(A) $p(x) = (x + 2)(x - 1)^2$

(B) $p(x) = -(x - 2)(x + 1)^2$

(C) $p(x) = -(x + 2)^2(x - 1)$

(D) $p(x) = -(x + 2)(x - 1)^2$

22.

HARD · NO CALCULATOR NEEDED

The length of a rectangle is 3 inches more than its width. The diagonal of the rectangle is 15 inches. What is the area of the rectangle, in square inches?

(A) 90

(B) 96

(C) 108

(D) 120

23.

HARD · NO CALCULATOR NEEDED

For $x > 0$, which of the following is equivalent to $\frac{(27x^9)^{2/3}}{x^{-2}\sqrt{81x^4}}$?

(A) x^{10}

(B) x^6

(C) $2x^6$

(D) $9x^6$

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 + ax^2 + bx + c$, where a , b , and c are constants, has $x - 1$ and $x + 2$ as factors. When $p(x)$ is divided by $x - 3$, the remainder is 40. What is the value of $p(4)$?

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

For which of the following values of k does the equation $\frac{2x}{x-4} - \frac{k}{x-4} = 3$ have no solution?

(A) 4

(B) 12

(C) -8

(D) 8

26.

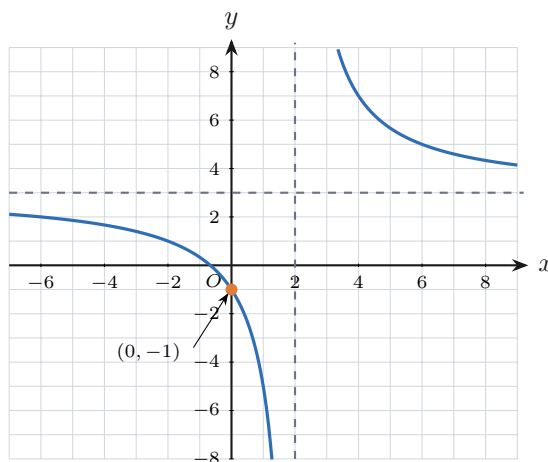
SAT CHALLENGE · NO CALCULATOR NEEDED

If $x > 0$ and $\sqrt{x} + \frac{1}{\sqrt{x}} = 5$, what is the value of $x + \frac{1}{x}$?

27.

SAT CHALLENGE · DESMOS

The graph of $f(x) = \frac{ax+b}{x-c}$ is shown in the xy -plane, where a , b , and c are constants. The dashed lines are the asymptotes of the graph, and the graph passes through $(0, -1)$. What is the value of $a + b + c$?



28.

SAT CHALLENGE · NO CALCULATOR NEEDED

For which of the following positive values of k does the equation $x^4 - 10x^2 + k = 0$ have exactly two distinct real solutions?

(A) 25

(B) 50

(C) 5

(D) 10

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

The positive quantities u , a , and b satisfy $\frac{1}{\sqrt{u}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}$. Which of the following gives u in terms of a and b ?

(A) $u = \frac{(\sqrt{a} + \sqrt{b})^2}{ab}$

(B) $u = \frac{ab}{(\sqrt{a} + \sqrt{b})^2}$

(C) $u = \frac{ab}{a + b}$

(D) $u = \frac{\sqrt{ab}}{\sqrt{a} + \sqrt{b}}$

30.

SAT CHALLENGE · CALCULATOR

A boat travels 24 kilometers upstream and then 36 kilometers downstream in a river whose current flows at 3 kilometers per hour. The entire trip takes 7 hours. What is the speed of the boat in still water, in kilometers per hour?

31.

SAT CHALLENGE • NO CALCULATOR NEEDED

For constants p, q and c , the equation

$$(2x - 3)(x^2 + px + q) = 2x^3 - 5x^2 + cx + 9$$

is true for all real values of x . What is the value of c ?

32.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a positive real number x , $\frac{\sqrt[3]{x} \cdot \sqrt[4]{x^3}}{\sqrt[6]{x}} = \sqrt[3]{k}$, where $k = 2^{11}$. Which of the following is equal to x ?

(A) $2^{\frac{121}{36}}$

(B) $2^{\frac{44}{3}}$

(C) 2^4

(D) $2^{\frac{11}{3}}$

33.

SAT CHALLENGE · NO CALCULATOR NEEDED

For $x > 0$ and any positive integer n , which of the following is equivalent to $\frac{(x^{2n+1})^3 + 4x^{6n+3} - 2x(x^{3n+1})^2}{x^{6n+1}}$?

(A) $3x^2$

(B) $5x^2$

(C) $3x^4$

(D) $3x^{12n+4}$

34.

SAT CHALLENGE · NO CALCULATOR NEEDED

For all $x > 1$,

$$\frac{a}{x-1} - \frac{bx+c}{x^2+x+1} = \frac{3x^2+8x+4}{x^3-1}$$

where a , b , and c are constants. What is the value of c ?

35.

SAT CHALLENGE · NO CALCULATOR NEEDED

For $y \neq 0$ and $y \neq \frac{1}{3}$, which of the following is equivalent to

$$\frac{\frac{1}{y^3} - 27}{\frac{1}{y} - 3} ?$$

(A) $\frac{1}{y^2} - \frac{3}{y} + 9$

(B) $\frac{1}{y^2} + 9$

(C) $\frac{1}{y^2} + \frac{6}{y} + 9$

(D) $\frac{1}{y^2} + \frac{3}{y} + 9$

36.

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 - 9x^2 + kx - 12$ has a positive integer constant k . For some positive integer n , $x - n$ is a factor of $p(x)$. What is the sum of all possible values of k ?

37.

SAT CHALLENGE · CALCULATOR

$$P = \frac{a}{\sqrt[5]{w^2}}$$

The formula above relates the positive quantities P and w , where a is a positive constant. Which of the following expresses w in terms of P and a ?

(A) $w = \left(\frac{a}{P}\right)^{2/5}$

(B) $w = \left(\frac{P}{a}\right)^{2/5}$

(C) $w = \left(\frac{a}{P}\right)^{5/2}$

(D) $w = \left(\frac{P}{a}\right)^{5/2}$

38.

SAT CHALLENGE · DESMOS

For a constant k , the equation

$$2\sqrt{x+1} + 2\sqrt{15-x} = k$$

has exactly one real solution. What is the value of k ?

(A) 8

(B) $8\sqrt{2}$

(C) 16

(D) $2 + 2\sqrt{15}$ **39.**

SAT CHALLENGE · NO CALCULATOR NEEDED

Positive real numbers x and y satisfy

$$\sqrt{x} - \sqrt{y} = 2 \quad \text{and} \quad x + y = 26.$$

What is the value of xy ?

40.

SAT CHALLENGE · NO CALCULATOR NEEDED

For how many integer values of k with $-10 \leq k \leq 10$ does the equation

$$\sqrt{x+k} = x-2$$

have exactly one real solution?

41.

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 + bx^2 + cx + 12$ has integer coefficients b and c , and $x + 3$ is a factor of $p(x)$. If p has exactly one real zero, what is the greatest possible value of c ?

42.

SAT CHALLENGE • NO CALCULATOR NEEDED

The polynomial $6x^4 + kx^2 + 24$, where k is a constant, can be written as $(2x^2 + a)(3x^2 + b)$, where a and b are integers. What is the greatest possible value of k ?

43.

SAT CHALLENGE • NO CALCULATOR NEEDED

For a real number a with $|a| \neq 1$,

$$\frac{1}{1-a} + \frac{1}{1+a} + \frac{2}{1+a^2} + \frac{4}{1+a^4} = 10.$$

What is the value of a^8 ?

- (A) $\frac{4}{5}$ (B) $\frac{5}{4}$
(C) $\frac{9}{5}$ (D) $\frac{1}{5}$

44.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a real number k , consider the equation

$$x - 4\sqrt{x} = k.$$

For how many integer values of k from -10 to 10 , inclusive, does the equation have exactly one real solution x ?

45.

SAT CHALLENGE · NO CALCULATOR NEEDED

For real numbers a and b , $a^2 - b^2 = 36$. What is the value of $(3a + b)^2 - (a + 3b)^2$?

46.

SAT CHALLENGE · NO CALCULATOR NEEDED

Let t be a positive real number with $t \neq 1$. Define $m = \sqrt[3]{t^2}$ and $n = \sqrt{t}$. If

$$\frac{m^p \cdot n^3}{\sqrt[6]{t}} = t^5,$$

what is the value of p ?

47.

SAT CHALLENGE · NO CALCULATOR NEEDED

The expression $\frac{(2x-3)^4(x+2)-7}{5}$ is equivalent to $ax^5 + bx^4 + cx^3 + dx^2 + ex + f$, where a , b , c , d , e and f are constants. What is the value of $a + b + c + d + e + f$?

(A) $-\frac{4}{5}$

(B) $\frac{3}{5}$

(C) -4

(D) 31

48.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$x^3 + (3 - k)x^2 - (10 + 3k)x + 10k = 0$$

In the equation above, k is a constant. The equation has exactly two distinct real solutions for exactly two values of k . What is the sum of these two values of k ?
